

Homework 4

Solution Key (Revenue-Share HHI, CR2, CR4)

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Definitions

Let firm i have revenue $R_i = p_i \cdot q_i$.

Let total market revenue be VM (column VM).

Define **revenue share** as:

$$sh_i = \frac{R_i}{VM} = \frac{p_i q_i}{VM}$$

- **Herfindahl–Hirschman Index (HHI, 0–10,000 scale)**

$$HHI = \sum_{i=1}^N (100 \cdot sh_i)^2$$

- **Concentration Ratio (CR_k)**

$$CR_k = \sum_{j=1}^k sh_{(j)} \quad \text{where } sh_{(1)} \geq sh_{(2)} \geq \dots$$

- **CR2:** sum of top two revenue shares
- **CR4:** sum of top four revenue shares

Use revenue shares for **all** calculations:

$$sh_i = \frac{p_i q_i}{V_M}.$$

Mapping to Excel Columns

Per month (row):

Revenues

- $r1 = p1 \cdot q1$ (Jeminy)

- $r2 = p2 \cdot q2$ (**Cousins**)
- $r3 = p3 \cdot q3$
- $r4 = p4 \cdot q4$
- $r5 = p5 \cdot q5$
- **All Others:**
 $r_others = \text{MAX}(0, VM - (r1 + r2 + r3 + r4 + r5))$

Shares

- $sh_i = r_i / VM$ for each of the 5 named firms (+ Others if $r_others > 0$)

Excel tips (replace `shares_range` with the cell range of shares):

- $HHI = \text{SUMPRODUCT}((100 \cdot \text{shares_range})^2)$
- $CR2 = \text{SUM}(\text{LARGE}(\text{shares_range}, \{1, 2\}))$
- $CR4 = \text{SUM}(\text{LARGE}(\text{shares_range}, \{1, 2, 3, 4\}))$

Do **not** use QM for HHI/CR — we are using **revenue** shares, not quantity shares.

Graphs Over Time

Create a Year–Month column and compute monthly **HHI**, **CR2**, and **CR4** using revenue shares. Produce three line charts: HHI, CR2, and CR4 vs. **date**.

Interpretation Thresholds

DOJ/FTC HHI screens (0–10,000 scale):

- **Unconcentrated:** $HHI < 1500$
- **Moderately concentrated:** 1500–2500
- **Highly concentrated:** > 2500

CR rules of thumb:

- **CR4** < 40% → fragmented / competitive
- **CR4** 40–60% → loose oligopoly
- **CR4** > 60% → tight oligopoly
- **High CR2** (> 40–50%) → strong duopoly tendency

Use both **levels** and **trends** to describe market structure.

Answering Management's Questions

- **HHI over time:** Report the level and trend. Compare to thresholds above.
- **CR2/CR4 over time:** Highlight concentration patterns. High/stable CRs indicate persistent market power.
- **Perfect competition?**
 - Only plausible if HHI is low (<1500) and CR4 < 40%.
 - Otherwise, the market is likely **oligopolistic**, not perfectly competitive.
- **Cousins' position:**
Report

$$sh_{\text{Cousins}} = \frac{p_2 q_2}{VM}$$

each month, note if it is a top-2 or top-4 firm, and whether its share is increasing or decreasing over time.

Common Pitfalls

- Forgetting **Others** when $VM > \sum(p_i q_i)$ → underestimates concentration.
- Mixing units (quantities instead of revenues).
- Shares must sum to 1 for accurate results.
- HHI/CR should always be calculated **monthly**, then graphed over time.